

Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition? Answer Choices: A. It activates TLR5 on the surface of macrophages. B. It primarily induces the Th1-cell response. C. Peptidoglycan is its major antigen that induces an immune response. D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<think> The patient's symptoms and treatment responses suggest that the patient has community-acquired pneumonia, which is commonly caused by bacteria such as Streptococcus pneumoniae. Cephalexin is a cephalosporin antibiotic that is effective against a variety of bacterial infections, including community-acquired pneumonia. The immune response elicited by a bacterial pathogen such as S. pneumoniae typically involves the activation of pattern recognition receptors on the surface of macrophages, such as Toll-like receptor 2 (TLR2) and TLR4, which recognize peptidoglycan and other bacterial components. This activation leads to the production of pro-inflammatory cytokines and the activation of the adaptive immune response. The Th1-cell response is primarily involved in the immune response to intracellular pathogens, such as viruses and some bacteria. The Th1 response is characterized by the production of cytokines such as IFN-gamma, which activates macrophages and enhances the elimination of intracellular pathogens. Encapsulation is a mechanism used by some bacteria, such as Streptococcus pneumoniae, to evade the immune system by forming a polysaccharide capsule that prevents phagocytosis. However, this is a characteristic of the pathogen, not the type of immune response it elicits. </think> <answer> C. Peptidoglycan is its major antigen that induces an immune response. </answer>

